

DSA with Python

Day 2

Arrays & Problem Solving

🎯 Goal:

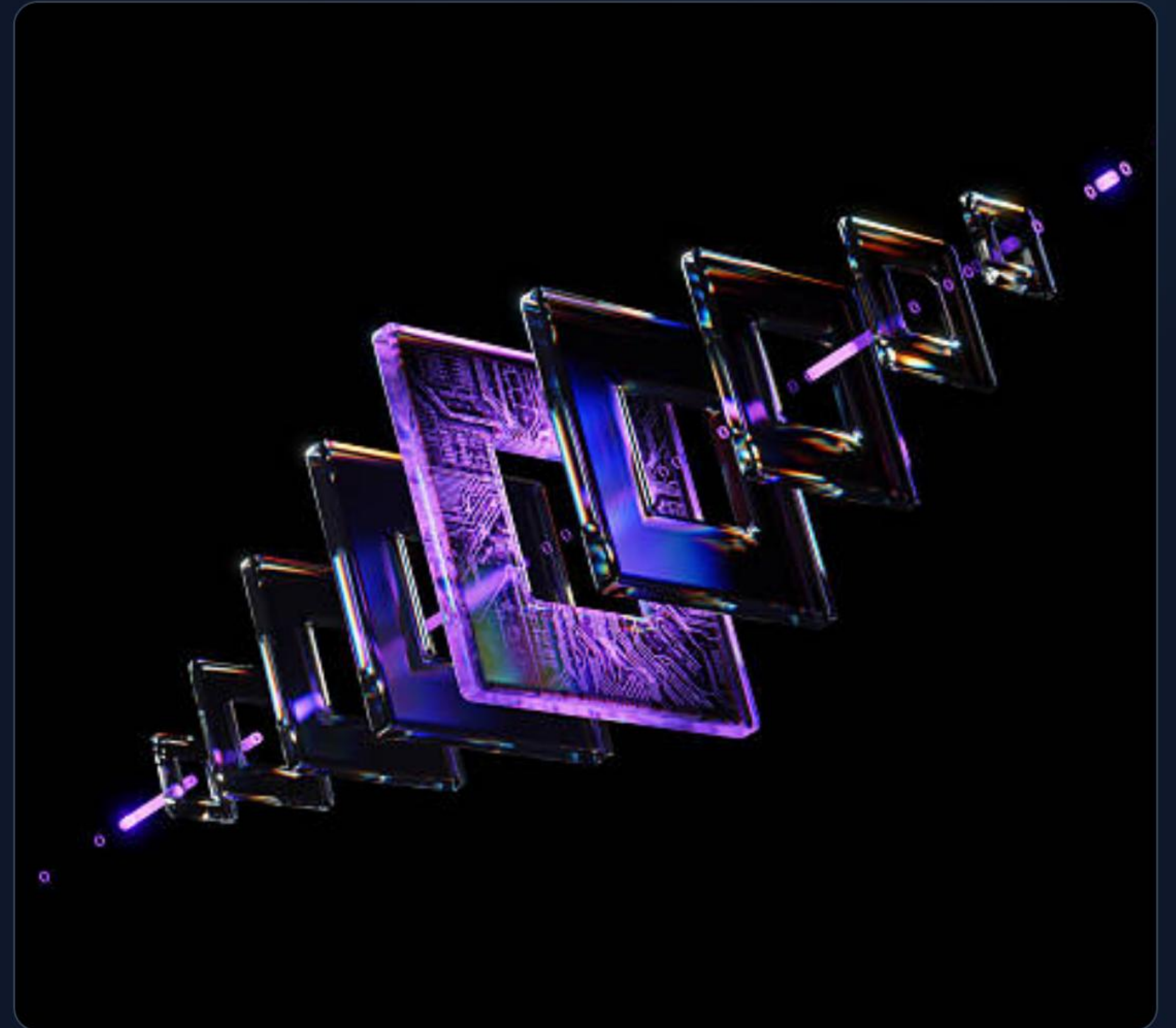
- Understand **Arrays** deeply
- Learn **Traversal**
- Solve basic problems
- Think like a problem solver 🧠

What is an **Array**?

- 📌 **Array** = Collection of values stored in continuous memory
- 📌 In Python → called **List**
- 📌 👉 Each element has a position (**index**)

```
numbers = [10, 20, 30, 40]
```

Index	0	1	2	3
Value	10	20	30	40



Real-Life Example:

Array



Think of a Train

Engine → Coach1 → Coach2 → Coach3

- Each coach = **element**
- Position matters strictly.
- 📍 Access any coach using its number (**index**).



Traversing an Array

 Traversing = Visiting each element one by one

✓ Method 1 (Best)

Directly iterating over the values. This is the most "Pythonic" approach.

```
for num in numbers:  
    print(num)
```

✓ Method 2 (Index Based)

Iterating using the length and index. Useful when position is required.

```
for i in range(len(numbers)):  
    print(numbers[i])
```

Real-Life Example: **Traversal**



Teacher checking students

-  Roll 1 → Check
-  Roll 2 → Check
-  Roll 3 → Check

 This sequential, one-by-one checking process is exactly what **traversal** looks like in code.

Problem 1: Sum of Array

</> The Logic

Example: [5, 10, 15, 20]

👉 Output: 50

```
total = 0
for num in numbers:
    total += num
```

💡 Real-Life Analogy

💰 Adding daily pocket money

Imagine putting your daily allowance into a piggy bank. You start with zero (`total = 0`), and every day you add the new amount (`total += num`).

Problem 2: Average

The Formula

$$\text{Average} = \frac{\text{Sum}}{\text{Number of elements}}$$

 **Real-Life:** Student marks → overall performance

The Code

Once you have the `total` from Problem 1, finding the average is just one division step away.

```
average = total / len(numbers)
```

Problem 3: Count Elements

len(numbers)

⚡ The Shortcut

The Manual Way

If we didn't have the Python built-in shortcut, we would use a traversal loop to count each item:

```
count = 0
for num in numbers:
    count += 1
```

Problem 4: Reverse Array

🧠 Real-Life: 🎥 Watching a video backward

✅ Method 1 (Loop)

Iterate backwards using the range function with a negative step.

```
for i in range(len(numbers)-1, -1,
-1):
    print(numbers[i])
```

✅ Method 2 (Shortcut)

Python list slicing makes reversing incredibly clean and easy.

```
numbers[::-1]
```

Slicing `[start:stop:step]` with a step of `-1` reverses the list.

Problem 5: Even Numbers

🔹 Filtering Logic

Use the modulo operator (%) to check if the remainder when divided by 2 is zero.

```
for num in numbers:  
    if num % 2 == 0:  
        print(num)
```

👉 Output:

2 4 6

Practice Questions

 Solve seriously:

 Sum of even numbers

 Find largest (without max)

Find smallest

 Count numbers > 10

  **Reverse without slicing** (Challenge)

2 Hour Plan

🕒 30 min

Dry Run

🕒 30 min

Concepts

🕒 1 hour

Practice



What's Next?



Day 3 Preview



Linear Search



Interview Logic



How search works (basic idea)

Image Sources



https://media.istockphoto.com/id/2213998103/photo/abstract-geometric-glass-elements-on-a-black-background-with-purple-reflections-digital-art.jpg?s=612x612&w=0&k=20&c=bm3-6pP3Jo41o-7ufae0zBH3QQ-Xd1q_5yOQy9kbbh4=

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